MT 1003 – Calculus and Analytical Geometry

Homework #8

Limits and Continuity of Functions of Several Variables

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Instructions:

1. Name the submitted file as `Section RollNumber HW8` (for example, `A 26I1234 HW8`).
2. Submit the complete homework as one scanned PDF file.
3. State the domain whenever a simplification, cancellation, or substitution depends on it. Unsupported final answers will not receive full credit.
4. To show that a limit does not exist, write each path explicitly and compute its limit. To show that a limit exists, give an argument that covers every approach: a limit law, a valid simplification, a bound, or an epsilon–delta proof. Agreement along a few paths is not a proof.
5. For Section B, include the Python code, generated output and graphs, and your written observations in the same PDF file.

Section A: Analytical Work

I. Evaluating Limits

Q1. Evaluate each limit. Name the method you use (factoring, rationalization, or a substitution $u = g(x, y)$), and state the domain where your simplification is valid.

(a) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^2+y^2}{\sqrt{x^2+y^2+9}-3}$

(b) $\displaystyle\lim_{(x,y)\to(2,0)} \frac{\ln(1+xy)}{y}$

(c) $\displaystyle\lim_{(x,y,z)\to(1,1,2)} \frac{x^2z-y^2z}{x-y}$

II. Limits That Do Not Exist

Q2. Use paths through the target point.

(a) Show that every straight line through the origin gives the limit 0 for $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^4y}{x^8+y^2}$; then find a curve that gives a different value.

(b) Show that $\displaystyle\lim_{(x,y)\to(1,-1)} \frac{(x-1)^2-(y+1)^2}{(x-1)^2+2(y+1)^2}$ does not exist.

(c) For $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x+y}{x^2+y^2}$, decide whether the limit is a real number, $+\infty$, $-\infty$, or none of these.

III. Proving That Limits Exist

Q3. Evaluate each limit. Justify your answer with the sandwich theorem or a one-variable substitution.

(a) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{\sin(x^2y)}{x^2+y^2}$

(b) $\displaystyle\lim_{(x,y)\to(0,0)} (x^2+y^2)\ln(x^2+y^2)$

Q4. Prove directly from the epsilon–delta definition that

$$\lim_{(x,y)\to(1,-2)} (x^2+2y) = -3.$$

Give an explicit formula for δ in terms of ε.

Q5. Use polar coordinates. For each limit, either prove that it exists with a bound independent of the angle, or show that it does not exist. In (b), identify the exceptional angles.

(a) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^2y^3}{x^4+y^4}$ (*Hint:* first show that $\cos^4\theta + \sin^4\theta \geq \frac{1}{2}$.)

(b) $\displaystyle\lim_{(x,y)\to(0,0)} \frac{x^2y^2}{x^2y^2+(x-y)^2}$

IV. Continuity

Q6. Decide whether each function is continuous at the origin. If it is discontinuous, classify the discontinuity as removable or non-removable, and for a removable one give the value that makes the function continuous.

(a) $g(x,y) = \begin{cases} \dfrac{x^2-2y^2}{x^2+y^2}, & (x,y)\neq(0,0), \\ 1, & (x,y)=(0,0); \end{cases}$ (b) $h(x,y) = \begin{cases} \dfrac{1-\cos(x^2+y^2)}{(x^2+y^2)^2}, & (x,y)\neq(0,0), \\ 1, & (x,y)=(0,0). \end{cases}$

Q7. Find the value of the constant a for which

$$f(x,y) = \begin{cases} a(x^2+y^2)+1, & x^2+y^2<1, \\ 4-(x^2+y^2), & x^2+y^2\geq 1 \end{cases}$$

is continuous on $\mathbb{R}^2$. For every other value of a, find exactly where f is discontinuous.

Q8. State whether each statement is true or false. Justify a true statement in one or two lines, and give a counterexample for a false one.

(a) If the limit of f along every straight line through $(0,0)$ equals L, then $\lim_{(x,y)\to(0,0)} f(x,y) = L$.

(b) If $|f(x,y)| \leq x^2+y^2$ for all $(x,y)\in\mathbb{R}^2$, then f is continuous at $(0,0)$.

(c) If $\lim_{(x,y)\to(a,b)} f(x,y)$ exists, then f must be defined at (a,b).

Section B: Exploring Limits with Python

Q9. Lines, Parabolas, and Level Curves. Run the following program for

$$f(x, y) = \frac{2x^2 y}{x^4 + y^2}, \qquad (x, y) \neq (0, 0).$$

The program prints the values of f at points approaching the origin along three lines and three parabolas, and then draws a contour map of f near the origin.

```python
import numpy as np
import matplotlib.pyplot as plt

def f(x, y):
    return 2*x**2*y / (x**4 + y**2)

t = np.array([0.1, 0.01, 0.001, 0.0001])

print("Along lines y = m x:")
for m in [1, 2, -1]:
    print("  m =", m, ":", f(t, m*t))

print("Along parabolas y = k x^2:")
for k in [1, 0.5, -1]:
    print("  k =", k, ":", f(t, k*t**2))

# Contour map (400 grid points per axis, so the origin is never a grid point)
x = np.linspace(-1, 1, 400)
y = np.linspace(-1, 1, 400)
X, Y = np.meshgrid(x, y)
Z = f(X, Y)

plt.figure(figsize=(6, 6))
cs = plt.contour(X, Y, Z, levels=np.linspace(-0.9, 0.9, 10), cmap="coolwarm")
plt.clabel(cs, fontsize=8)
plt.title("Level curves of f(x, y) = 2x^2 y / (x^4 + y^2)")
plt.gca().set_aspect("equal")
plt.show()
```

Using the output, the graph, and exact calculations, answer the following.

(a) From the printed values, what number do the values along the lines $y = mx$ appear to approach? Prove this exactly for every line through the origin, including the vertical line $x = 0$.

(b) What do the printed values along the parabolas $y = kx^2$ show? Prove that $f(t, kt^2) = \dfrac{2k}{1 + k^2}$ for $t \neq 0$.

(c) Does $\lim\limits_{(x,y)\to(0,0)} f(x, y)$ exist? Justify your answer using (a) and (b).

(d) Show that every level curve $f = c$ with $c \neq 0$ is made up of parabolas $y = kx^2$, and find which values of c actually occur. How does the contour map show this?

(e) Can any value assigned at $(0, 0)$ make f continuous there? Explain in one or two sentences.

(f) In two or three sentences, explain why the printed table alone could never prove that a limit exists, even if every row approached the same number.